01/24

Friday, 24 January 2025 9:30 AM

Direct Methods for $Ax = b$

+ Cramer's rule
+ Gaussian Elimination

  $Ax = b$ → converted to $Lx = b'$ or $Ux = b'$

+ Gauss-Jordon
+ LU decomposition

Variations of the same

When do you not need to do this?
↳ Diagonally dominant & positive definite

+ Full & partial pivoting [moving rows & columns]
+ Equilibration → normalize all eqn so you can compare the eqn
  → essential to do before pivoting

(direct)
These methods are memory intensive
(i.e. all the elements (even zero) need to be stored ↳ since they may even become non-zero)

Thomas Algorithm

Multiplying two diagonal matrices → gives tridiagonal matrix

Iterative Methods

+ Jacobi (parallelizable)
+ Gauss Seide (cannot be parallelized)

guaranteed always converge if A is diagonally dominant

How to check?
+ rearrange eqn so you have largest terms in diagonal
+ solve for just the diagonal elements

For linear systems → initial guess does not matter
↳ If no converge → successive over/under relaxation (SOR)
↳ introduce ω parameter

★

$Ax = b$ error analysis

+ Residuals
  ↳ But a small residual only makes sense if the matrix is ill-conditioned

Always do relative error!

$$\kappa(A) \geq 1$$

$\kappa(A)$ expensive to calculate

$$\kappa(A) = \|A^{-1}\| \|A\|$$

$$\frac{\|\delta x\|}{\|(x + \delta x)\|} \leq \kappa(A) \frac{\|\delta A\|}{\|A\|}$$

+ Iterative Refinement Algorithm
  ↳ Calculate residual at twice the precision (significant figures)

(only one iteration p much needed)